TBdecompile Reference Manual

Contents

ToolBook II 8.1 — Operational Notes

15 October 2026

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Appendices are in the companion volume.

Object Create Path

Source: 20261012/01-object-create-path.md

Object create path and bounds

From the honest unpack (`tbwp0001/TB80BAS_noghidra.DLL`) loaded in Ghidra.
Source writeup: `tbwp0002/bas-create-path.md`.

Object layout

• Object size is **0x76** bytes. `CDBNEWOBJECT` indexes them as `param_6 * 0x76`.
• Bounds are four words at **object+0xf, +0x11, +0x13, +0x15** (left, top, right, bottom).
• Flags byte at object+0x1f. Bits set from the source structure at +0x0a, +0x0c, +0x0e.
• Source-structure fields copied by `FUN_12c8_0ef8`:
- source+0x3e / +0x40 / +0x42 / +0x44 → object+0x17 / +0x19 / +0x1b / +0x1d
- source+0x06 → object byte at offset given by a type table
- source+0x04 → pattern lock

Create path

```
CDBSENDCREATE          (12c8:00bc, Ordinal_51)
  ■■ FUN_12c8_0850     (12c8:0850)   one caller only
       ■■ FUN_12c8_0e32 (12c8:0e32)  allocator
            ■■ FUN_12c8_0ef8 (12c8:0ef8)  field copier / bounds writer
```

`FUN_12c8_0850` is not the record reader. It locks context, calls the allocator, and tears the object down on error. `ENTER 0x4,0x0`.

CDBSENDCREATE type dispatch

`param_5` is the type code. The string arithmetic in the decompile is Ghidra matching constants against `s_CTbkSimpleOLEDispatch_CallMethod`.

param_5	path
0x1014, 0x1015	`FUN_12c8_0850` — new object
0x1021, 0x1022	`FUN_12c8_0a86`
0x3018–0x301b	bounds-checked with `CDBCHECKBOUNDDTL`, then `FUN_12c8_0e08`
0x3022	`FUN_12c8_0d40`
else	`Ordinal_115`

Parameters: `param_3`/`param_4` are the bounds pair (clamped against 32000 and 0x8300). `param_6`/`param_7` are compared to the current object id. `param_9` is an error out-byte.

FUN_12c8_0ef8 bounds write

Two paths write the four bounds words:

1. When `param_2` is null, copy from globals set from `param_4`/`param_5`:

```
object+0x0f = DAT_1630_01fe;  // param_4
object+0x11 = DAT_1630_0200;  // param_5
object+0x13 = DAT_1630_0202;  // param_4
object+0x15 = DAT_1630_0204;  // param_5
```

2. In the `switch(param_6)` default case, copy four words from `param_2` into the same offsets.

`param_6` is the object type:

```
| param_6 | path |
|---------|------|
| 0x0f, 0x10, 0x13, 0x14 | polygon-like; `OBJNEWNEAR(param_1 << 2)` for a point array |
| 0x11, 0x12 | break (arc/pie; 0x11 matched the SoftICE arc) |
| 0x24 | special; also copies four words from `param_2` |
| default | copies `param_2[0..3]` into object+0xf..+0x15 |
```

Property table

`FUN_1058_0a98` at **1058:0a98**. Builds the property-id list.
First write of `0x4004` is at decompile line 30313.
Nine writes of `0x4004` in the function.

Ids written: `0x4004` (bounds), `0x4009`, `0x400d`, `0x4015` (idNumber), `0x401a`, `0x401b`, `0x401c`, `0x4020`–`0x4022`, `0x4023` (script), `0x4027`, `0x402b`, `0x4031`–`0x4033`, `0x4034` (vertices), `0x4035`.

Still open

`CDBSENDCREATE` receives bounds as parameters. It does not parse a `14 00 00` record. The record-to-bounds reader is a caller of `CDBSENDCREATE`, either in BAS or in the book-open path in RTM/Tbload.

Next step: list XREFs on `CDBSENDCREATE`. The caller that passes a pointer into a page blob is the reader.

Graphics Painters

Source: 20261012/02-graphics-painters.md

Graphics painters (TB80BAS)

Confirmed from the decompile and SoftICE traces.

Painters

```
| Function | Segment:offset | Role | Notes |
|----------|----------------|------|-------|
| `FUN_1408_0376` | 1408:0376 | PaintPolygon | Count at object+0x30. Calls `SetupPaintRect`, `ScaleTwipsToScreen` on DS:30AC, then `GD
| `FUN_1410_0066` | 1410:0066 | PaintRectangle | Reads bounds at object+0xF, calls `GDI!RECTANGLE`. |
| `FUN_1008_01dd` | 1008:01dd | ScaleTwipsToScreen | `screen = (stored * DPI + 720) / 1440`. At DPI 96: stored/15. |
| `FUN_10a0_0558` / `FUN_10a0_0458` | 10a0 | Arc helpers | Live object type 0x11, four twip pairs, fill at +0x18. |
| `FUN_1408_00d8` | 1408:00d8 | SetPointArray | Runtime property 0x4050 setter. Not on disk. |
```

On-disk polygon (solved)

- Signature: `14 00 00` subtype-0, `+0x0c = 04 00`.
- Fill dword at +4.
- Point count at +0x1A.
- x,y twip pairs start at +0x22.
- Screen = twips/15.
- 96 such records on 13 pages in 9e954d01; P430 has 22.

Still open

```
| Item | Status |
|------|--------|
| Curve control structure (P420) | Points exist at page+0xA13. No count/header recovered. Render is a tangle. |
| Arc disk layout | Live object known. No matching `14 00 00` pattern tied on disk. |
| Picture id → DIB | 641 BMPs carved. Widget field that holds the id not identified. TB80BMP ordinal 3 is the reader. |
| Computed shapes (brackets) | Segment 0x5B builds them from bounds. Not stored points. |
```

Scale function (recovered)

```
// FUN_1008_01dd — simplified
// For each x,y pair:
//   screen_x = (stored_x * DAT_1630_01a6 + 0x2d0) / 0x5a0
//   screen_y = (stored_y * DAT_1630_01a8 + 0x2d0) / 0x5a0
// DAT_1630_01a6 / 01a8 = 96 on the observed boot → /15.
```

OpenScript and System Books

Source: 20261012/03-openscript-and-system.md

OpenScript and NIDA system books

OpenScript decoder

• Keyword table is resource 0x77 (type 100, id 119). Offset table at byte 2 + count*4.
• High-bit opcodes (`0xE0`, `0xF0`, `0xFF`, …) emit whitespace, CR/LF, or repeats.
• Back-references (`0xC0`–`0xFF`) copy from the compressed input stream.
• Decoder stops at the first standalone `end`.
• Handles `go to page`, `send next`/`previous`, basic conditionals, and system properties.
• Corpus: 76 handlers from 9e954d01.tbk, 100 buttonClick handlers from nida81.tbk.

tb80r.sbk compression

• Uses the same bytecode as the lesson books.
• Difference: the handler name prefix ("to handle" / "to get") is also compressed.
• 33 handlers extracted (`tbwp0004/tb80r_handlers.json`).
• Encoding rules are identical; byte distribution differs (more 0xFF/0xE0).

NIDA system books

```
| File | Role |
|------|------|
| `nida81.tbk` | Shared shell. 43 objects, pages P20000–P22130. 100 buttonClick handlers. |
| `nc81base.sbk` | Resource container. Object-type table. No OpenScript. |
| `nc81recs.sbk` | Resource container. No OpenScript. |
| `nc81test.sbk` | Resource container. No OpenScript. |
| `nc81trnr.sbk` | Resource container. No OpenScript. |
| `nc81int.dll` | 47 exports. Trainer/comms (IntCheckComm*, voltages, switches, faults). |
| `nc81rec.dll` | Session recording (RecStartSession, RecWriteResponse, RecBookmarkSession, …). |
```

Lessons call `initializebase`, `initializerecs`, `initializetest`, `openSession`, `gotoPageList`.

These resolve to nida81.tbk handlers or the two DLL exports. The `.sbk` files hold no script.

Still open

• Full call inventory across the ATT Master_3 set (not just one lesson).
• Minimal stubs for the closed NIDA call set.
• Picture-id binding and remaining draw-object disk layouts.

Symbol Maps

Source: 20261012/04-symbol-maps.md

Symbol maps

Generated from the Ghidra decompiles. Source: `tbwp0003/`.

File	Entries	Source
`TB80BAS.map`	3799	TB80BAS_noghidra.DLL.c
`TB80RTM.map`	2669	TB80RTM.EXE.c

Format

```
SSSS:OOOO name
```

`SSSS` is the NE segment index from the decompile (for example `13D0`). `OOOO` is the offset inside that segment. This is not a live selector. The selector changes every boot. Match the segment index to the live selector in the debugger, then use the offset.

Named exports (`CDBSENDCREATE`, `Ordinal_51`, etc.) replace the `FUN_` name when the decompile attached one. Everything else keeps the Ghidra name.

SoftICE

SoftICE does not load this MAP directly.

1. Break on the BAS ordinal and walk back (no symbols needed):

```
EXP TB80BAS
BPX TB80BAS.Ordinal_51
```

2. Or convert the MAP to a SYM if you have a converter.

Do not use `BAS(82)` or similar. That is not a SoftICE address. Use `MAP TB80BAS`, read the selector for the segment, then `BPX :offset`.

WDW

WDW does not load this MAP as Watcom debug info. Use it as a lookup:

1. Note the live selector for the NE segment.
2. Find the function in the MAP.
3. Set the breakpoint at `selector:offset`.

Or break on BAS ordinal 51 and read the return address.

Limits

- Offsets are from the unpacked image. They match the decompile.
- No types, no locals, no line numbers. Names and addresses only.
- The segment index must be matched to the live selector by hand.

Open Questions

Source: 20261012/05-open-questions.md

Open questions

Bounds

• Disk `14 00 00` record → object+0xF expansion rule.
• `CDBSENDCREATE` receives bounds as parameters. The reader is a caller.
• Next: XREFs on `CDBSENDCREATE` in BAS or the RTM/Tbload open path.

Graphics

• Curve control structure (P420 flame). Points exist; order/structure wrong.
• Arc disk layout. Live object (type 0x11) known; disk match missing.
• Picture id → DIB binding. BMPs carved; widget field unknown.
• Computed shapes (brackets, lines) once bounds exist.

OpenScript / system

• Full NIDA call inventory (initialize*, session, gotoPageList).
• Stubs for the closed set that appears in the corpus.
• Modern CMP expander (standalone, no W16). Test corpus already extracted.

Engine

• RTM book-open path. Unicorn hosts die before a book open (FFFF:0158 / unbound far call).
• CMP W16 segment-not-present fault (001c:00000015). Needs live selector capture.
• TBKX export via ToolBook 11.5. Prospective; no samples in tree yet.

Closed

• FIDB trial. moralrecordings DOS/Win16 databases produce zero CRT renames.
• Compiler is Microsoft C (R60xx strings). VC++ 1.52c / MSC 7.0 already covered.
• SoftICE `BAS(nn)` syntax does not work. Use MAP → selector.

Debugger Notes

Source: 20261012/06-debugger-notes.md

Debugger notes (SoftICE / WDW)

From field notes and the corrected breakpoint discussion.

SoftICE standing orders

• `BC *` before the next breakpoint. A live `GDI!POLYGON` breakpoint freezes SoftICE if the page scrolls.
• Selectors change every boot. Copy offsets. Remap selectors.
• `MAP TB80BAS` (or RTM, BMP, …) to get the live selector for a segment.
• Segment numbers in MAP output are hex.
• Do not search all of memory. Wide 80 65 searches can GPF the VM.
• A dump that is the SoftICE window or the DOS device chain (CON/AUX/PRN) missed.

Correct breakpoint form

```
BC *
MAP TB80BAS
; read the selector for the segment you want (e.g. 0x82)
BPX <selector>:0376 DO "D SS:BP-4 L 4; D DS:30AC L 40"
G
```

Invalid: `BPX BAS(82):0376`

SoftICE does not accept that notation.

Useful offsets (stable)

```
| Function | Segment:offset | Notes |
|----------|----------------|-------|
| PaintPolygon | 1408:0376 | Object at [BP+04], count at +0x30, points at +0x32 |
| PaintRectangle | 1410:0066 | Bounds at object+0xF |
| ScaleTwipsToScreen | 1008:01dd | |
| CDBSENDCREATE | 12c8:00bc | Ordinal_51 |
| FUN_12c8_0850 | 12c8:0850 | Allocator caller |
| FUN_12c8_0ef8 | 12c8:0ef8 | Bounds writer |
```

WDW

• Use the `.map` files as a lookup. Note the live selector, then set `selector:offset`.
• Or break on BAS ordinal 51 and read the return address to land in the caller.

Dump cautions

• Real dumps: page.bin01511 (ASYM_TpID), p400.bin0157 (JBO signature), dump-001/002 (Viewer/System).
• Junk: console logs, DOS device chain, name-slot only.
• ICEDUMP 6.021 needs kernel32 and user32 exports in winice.dat. Memory-dump recipe is not repeatable.